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College of Engineering
EE 456 RF & mm-Wave Active Circuits

Design Project 3

Matching Network Synthesis Techniques Using Chebyshev Polynomials

Design Project #2

Element	Score	Max
Design		300
Input Matching Network		100
Output Matching Network		100
Simulations (MATLAB)		200
Simulations (ADS)		100
Comparison		100
Presentation		100
Overall		1000

Design Project Objective

Employing NEC's NE321000 Ultra Low Noise Pseudomorphic HJ FET ($V_{DS} = 2$ V and $I_{DS} = 10$ mA) design an amplifier that operates from 10 to 20 GHz ($f_L = 10$ GHz & $f_H = 20$ GHz) with a gain of approximately 8.15 dB ± 0.17 dB.

Design the input matching network to operate from 9 GHz to 20 GHz ($f_L = 9$ GHz & $f_H = 20$ GHz). The matching network should employ seven reactive elements and be designed for a sloped response (6 dB/octave). Lastly, assume that $IL_{\min(\text{dB})} = 0.0$ dB and Ripple $= 0.1$ dB.

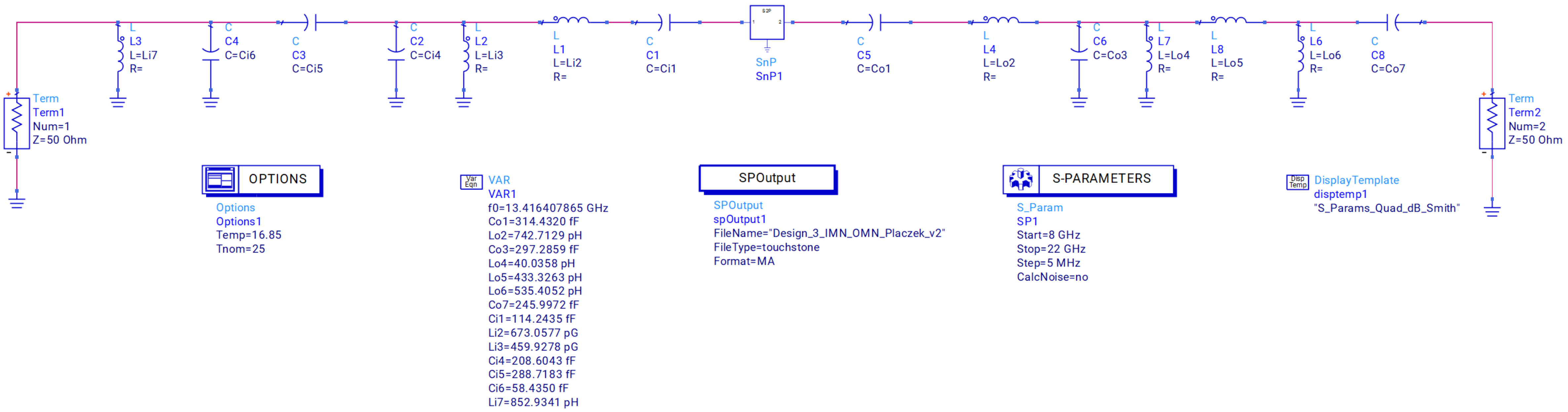
Design the output matching network to operate from 9 GHz to 20 GHz ($f_L = 9$ GHz & $f_H = 20$ GHz). The matching network should employ seven reactive elements and be designed for a non-sloped response. Lastly, assume that $IL_{\min(\text{dB})} = 0.1$ dB and Ripple $= 0.1$ dB..

Design and Simulation Results

Parameter	Value	Unit	Connection
Z0	50	Ω	Termination
Li7	852.9341	pH	Shunt
Ci6	58.435	fF	Shunt
Ci5	288.7183	fF	Series
Ci4	208.6043	fF	Shunt
Li3	459.9278	pH	Shunt
Li2	673.0577	pH	Series
Ci1			Series
FET Gate			
FET Drain			
Co1			Series
Lo2	742.7129	pH	Series
Co3	297.2859	fF	Shunt
Lo4	40.0358	pH	Shunt
Lo5	433.3263	pH	Series
Lo6	535.4052	pH	Shunt
Co7	245.9962	fF	Series
Z0	50	Ω	Termination

Table 1

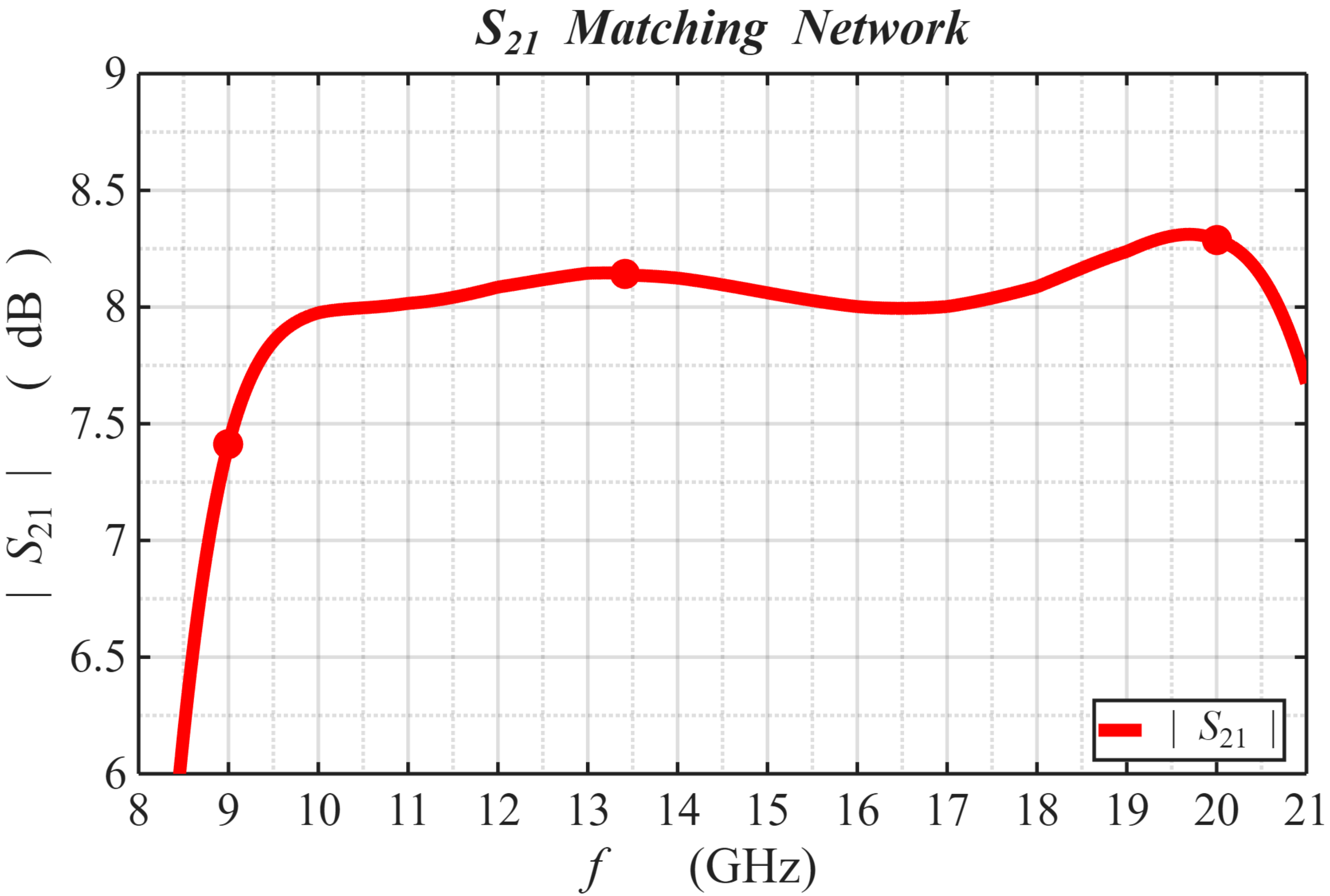
ADS Schematic of Design



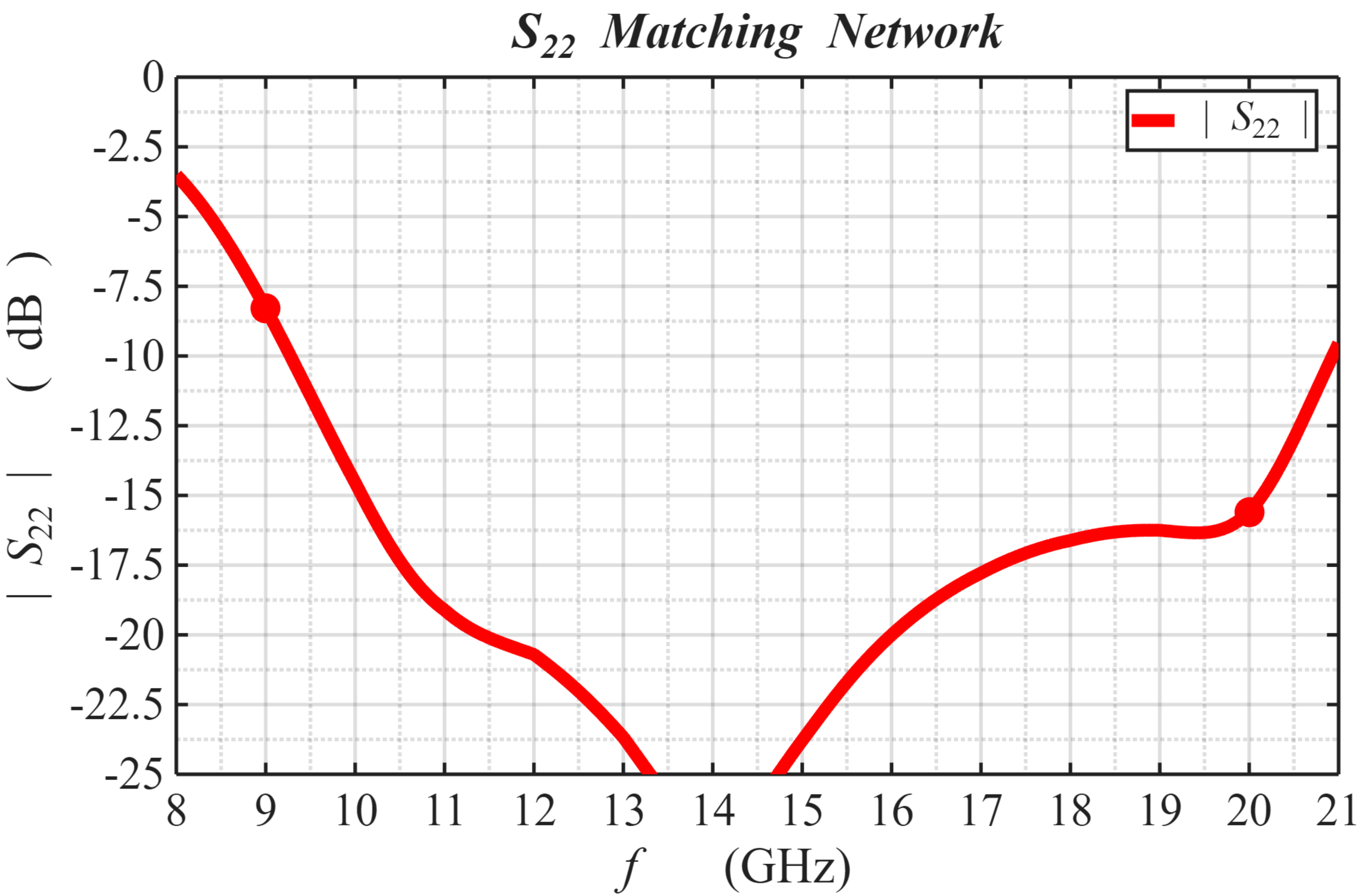
Design Verification

ADS & MATLAB Simulations

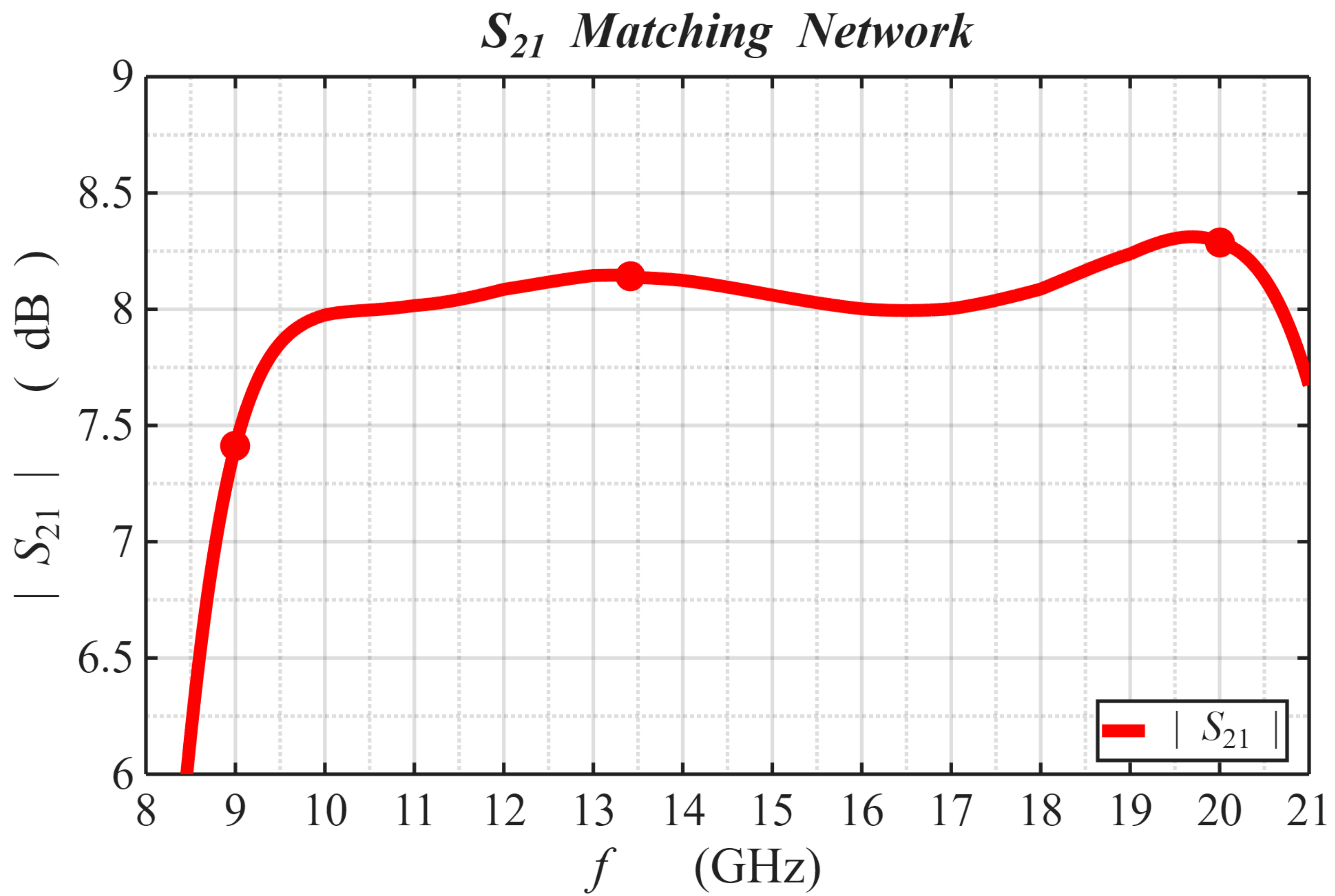
Matlab Simulation



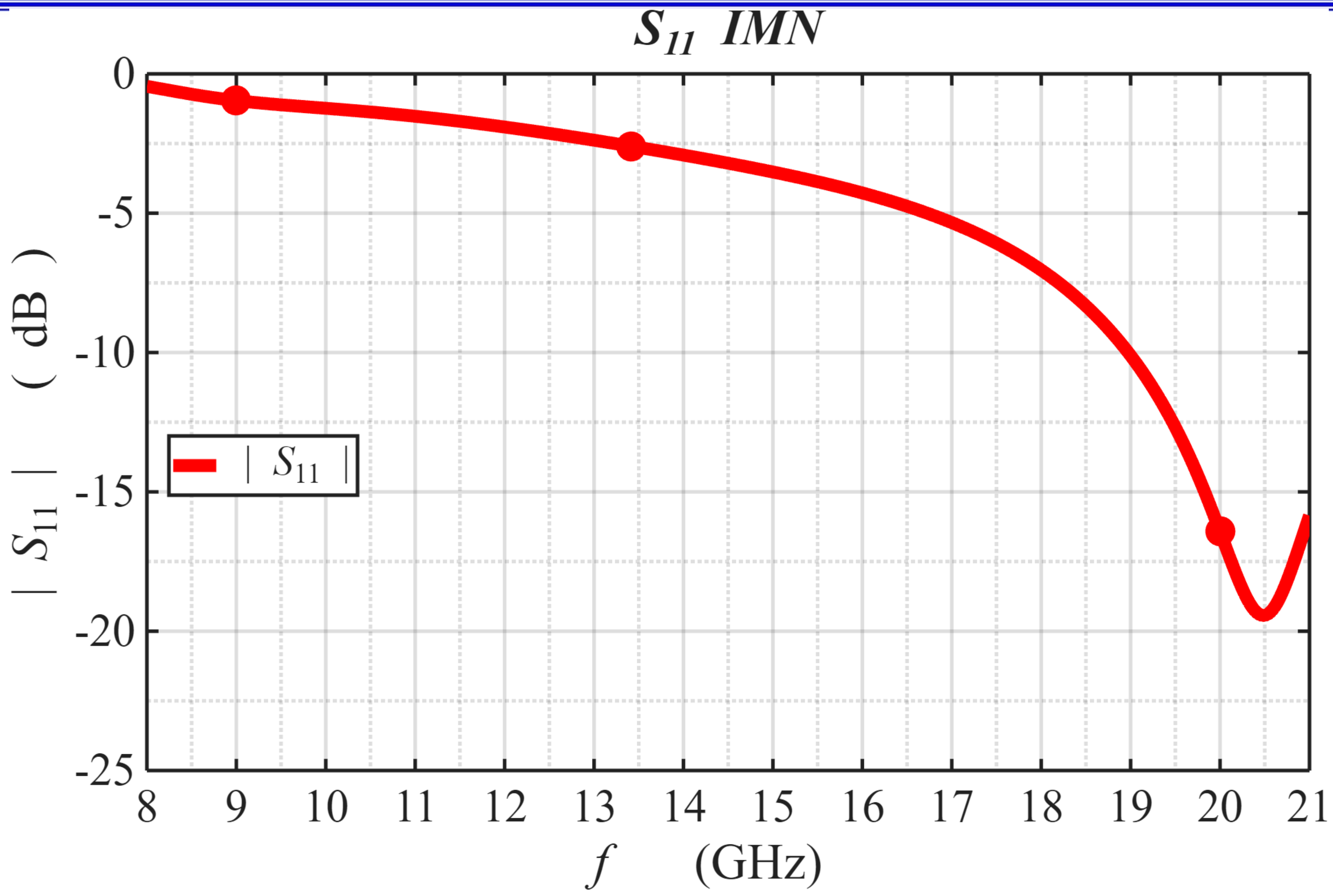
Matlab Simulation



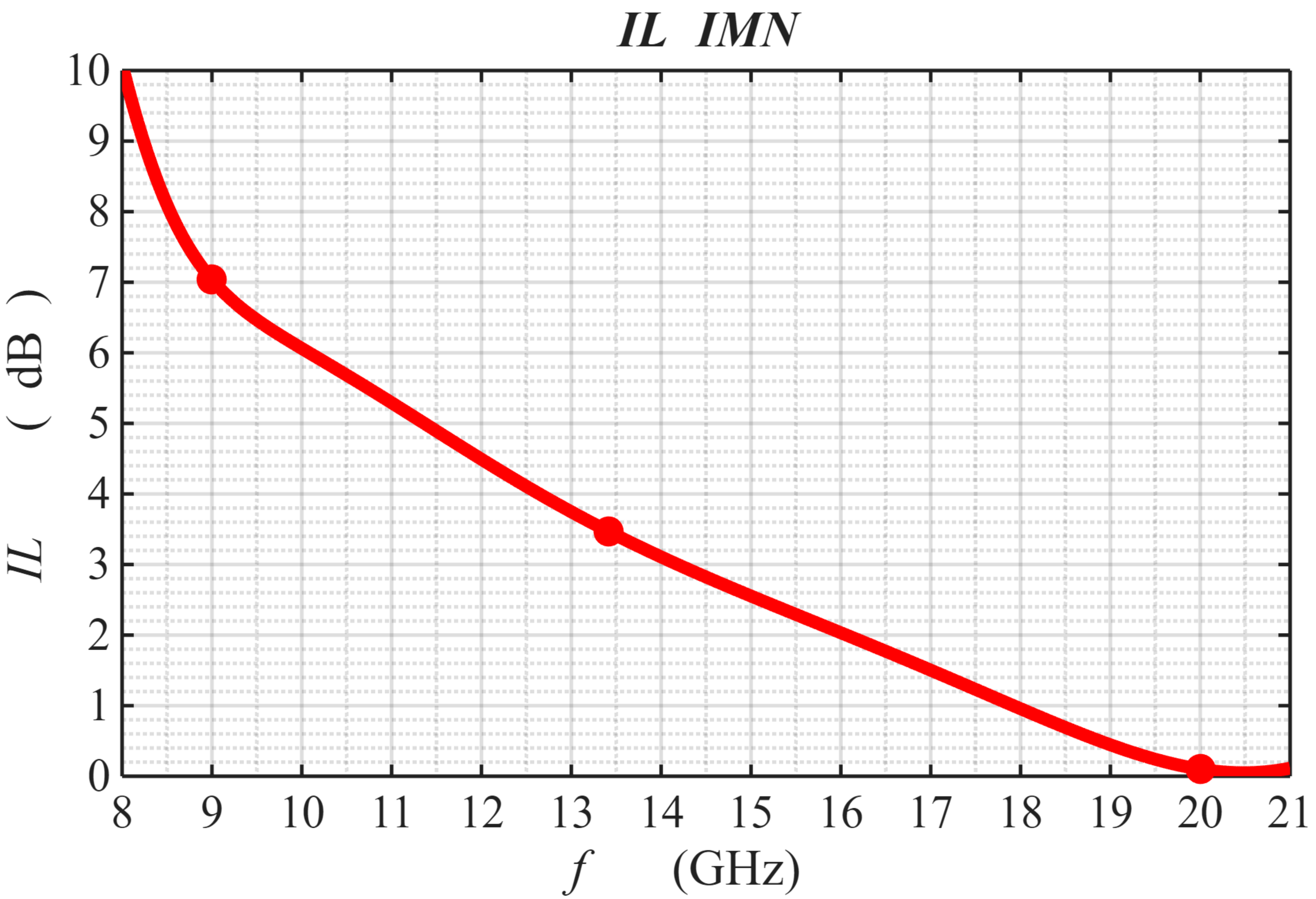
Matlab Simulation



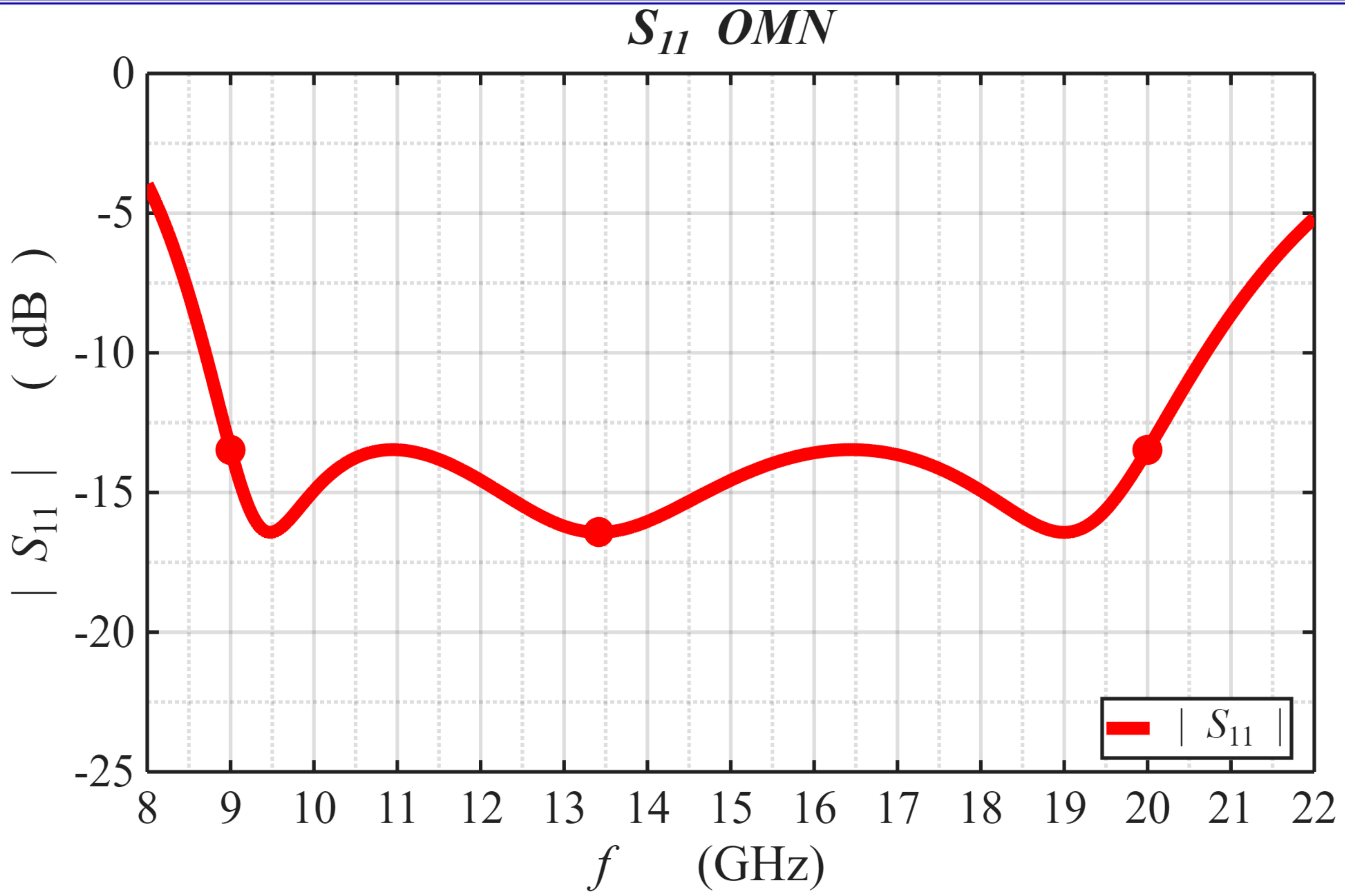
Matlab Simulation



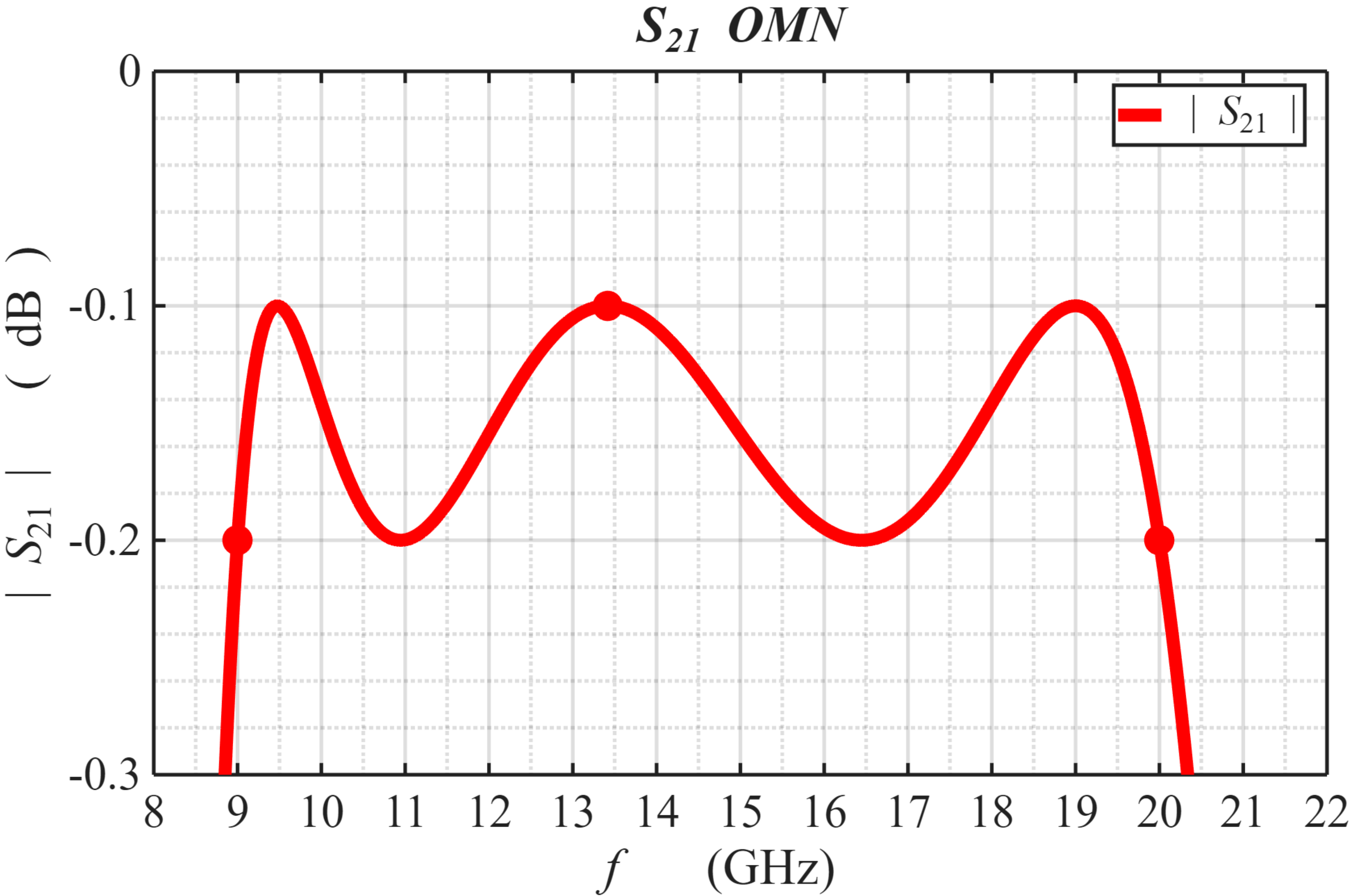
Matlab Simulation



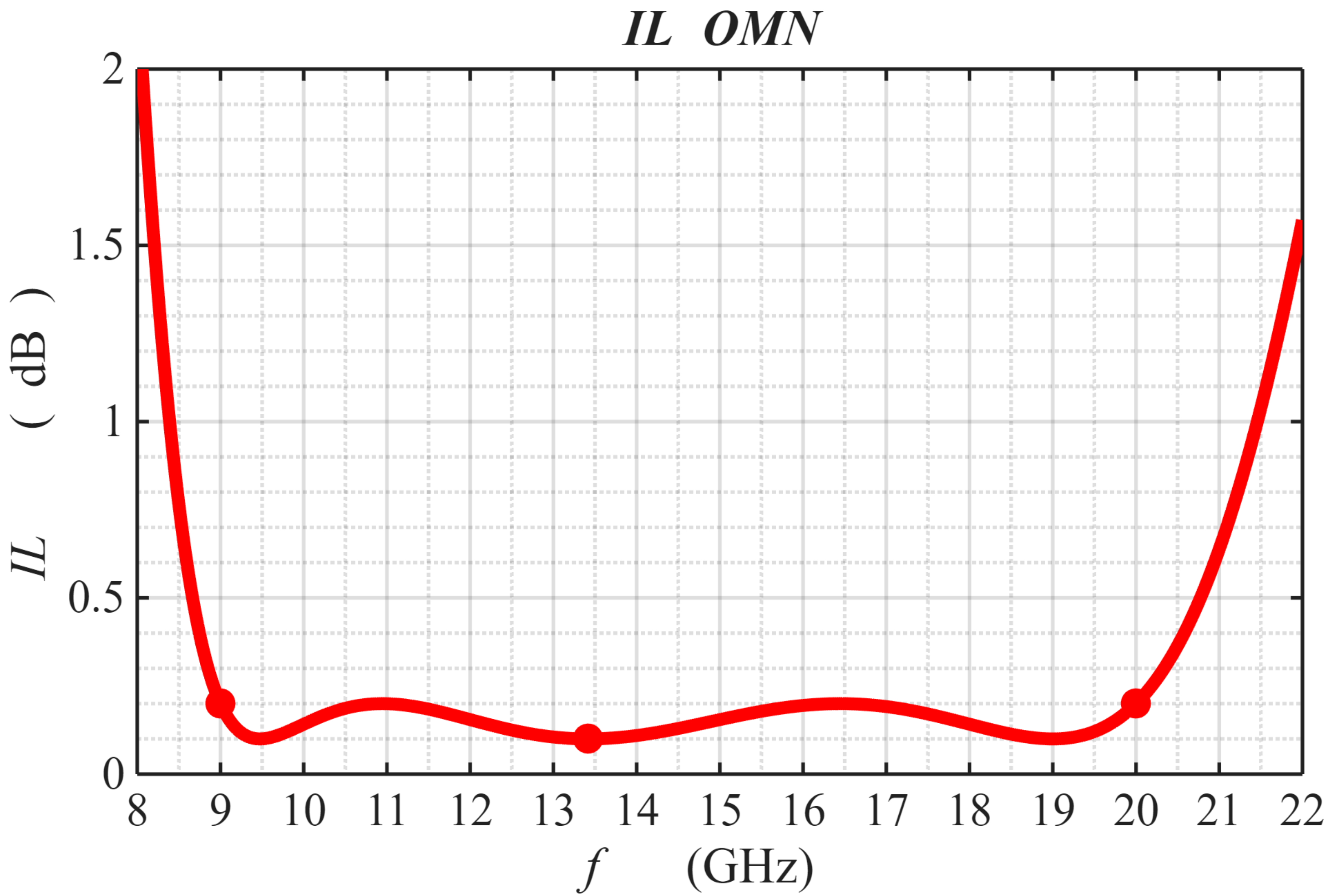
Matlab Simulation



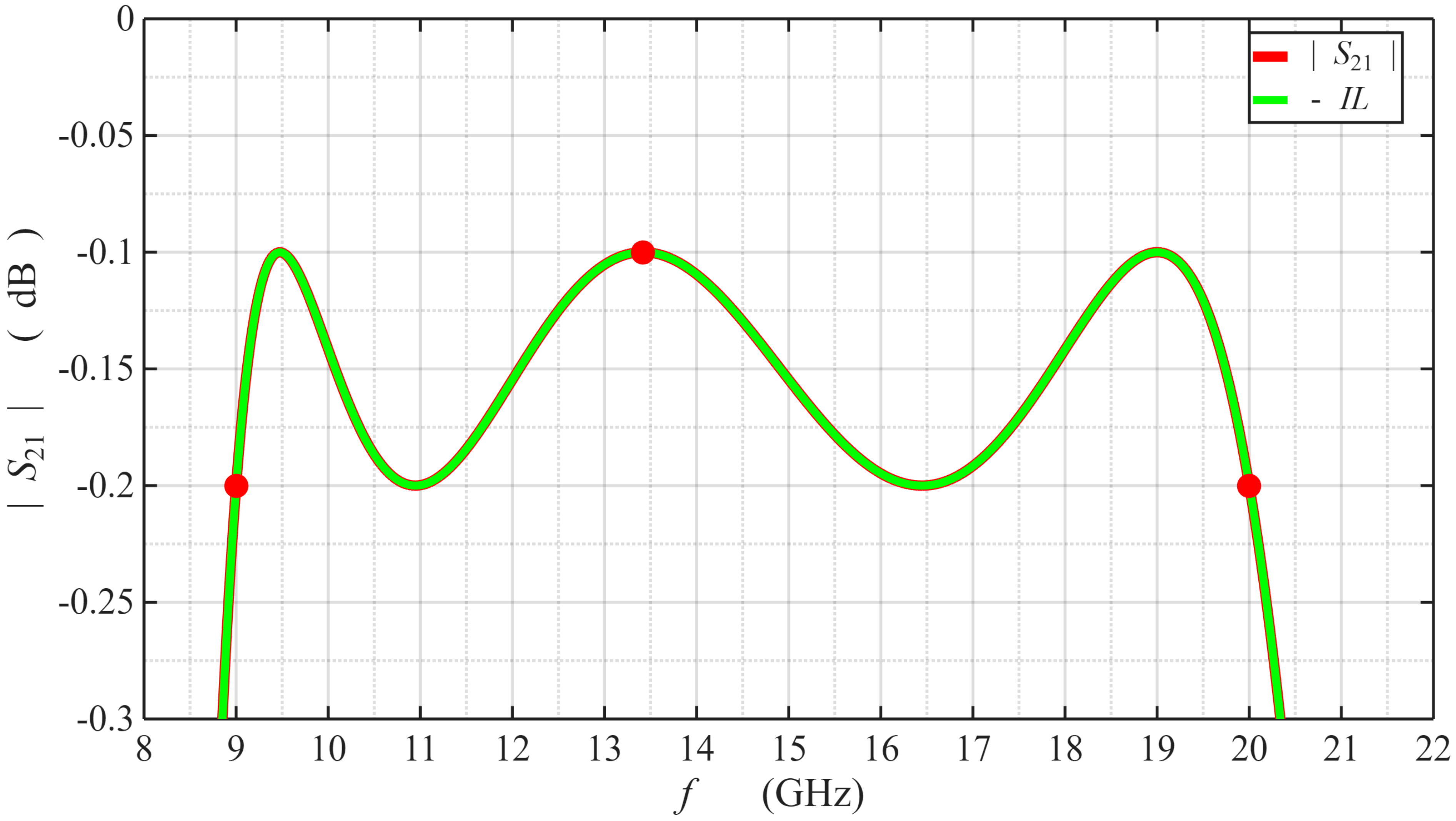
Matlab Simulation



Matlab Simulation



Matlab vs ADS Simulations



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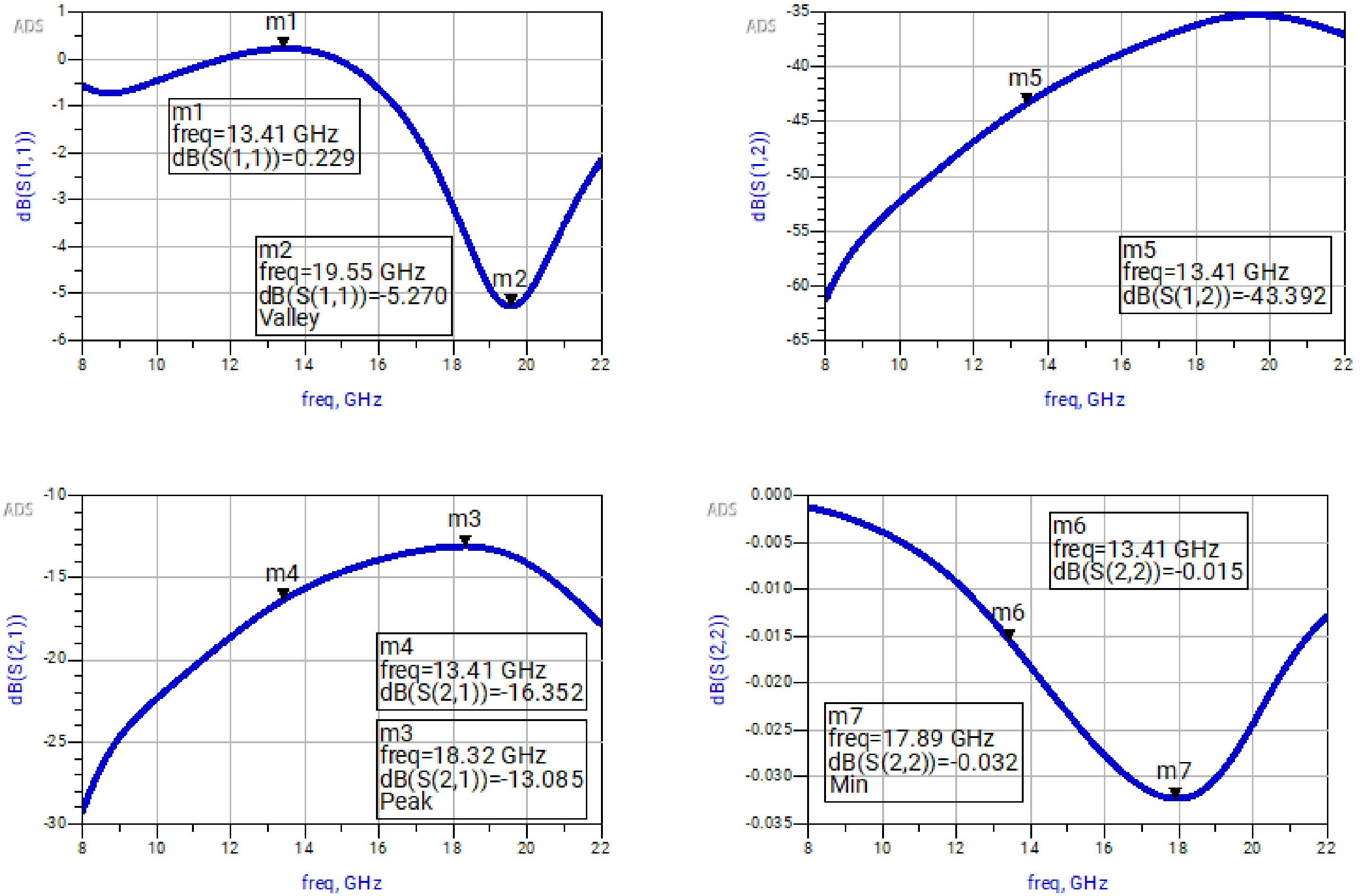
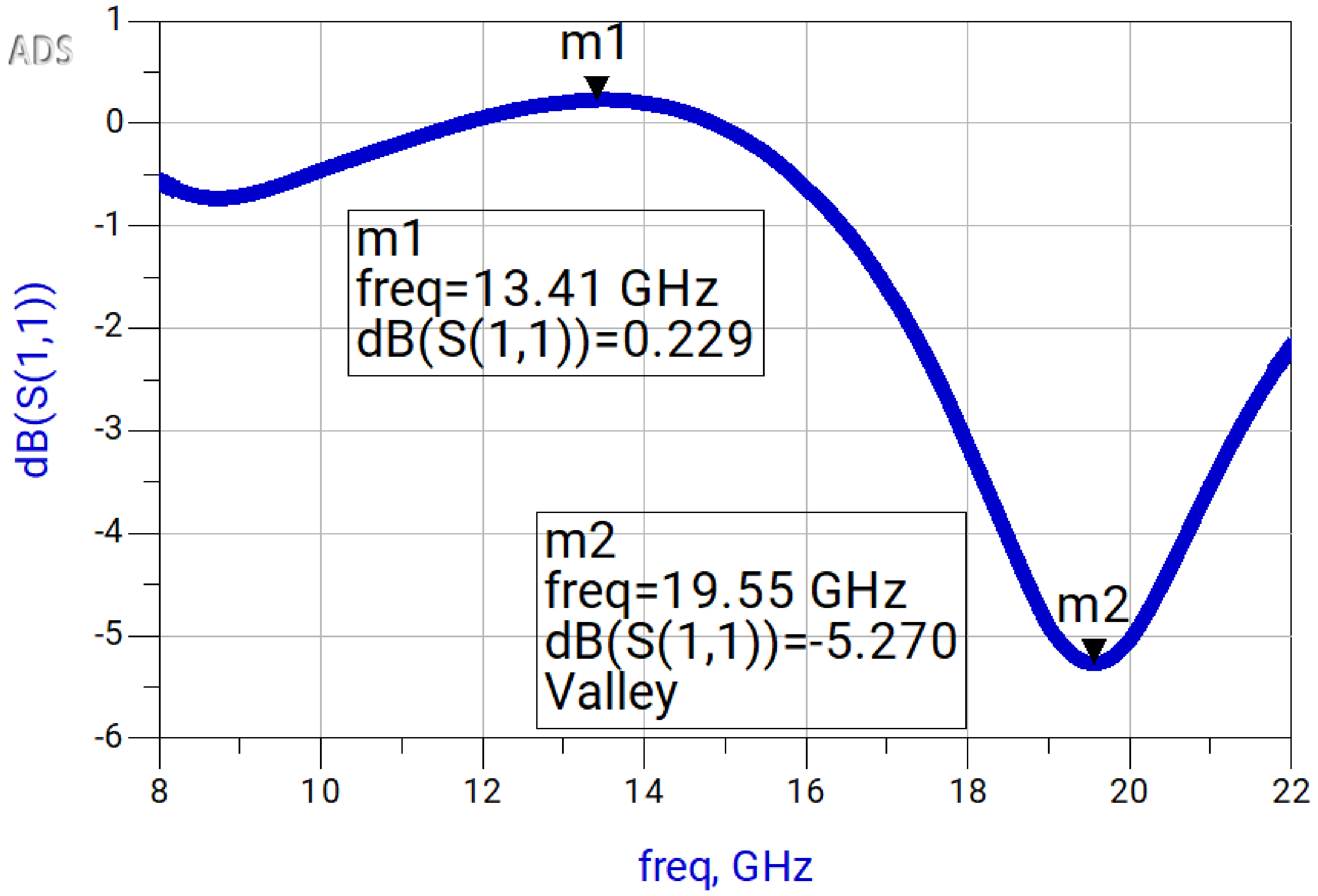
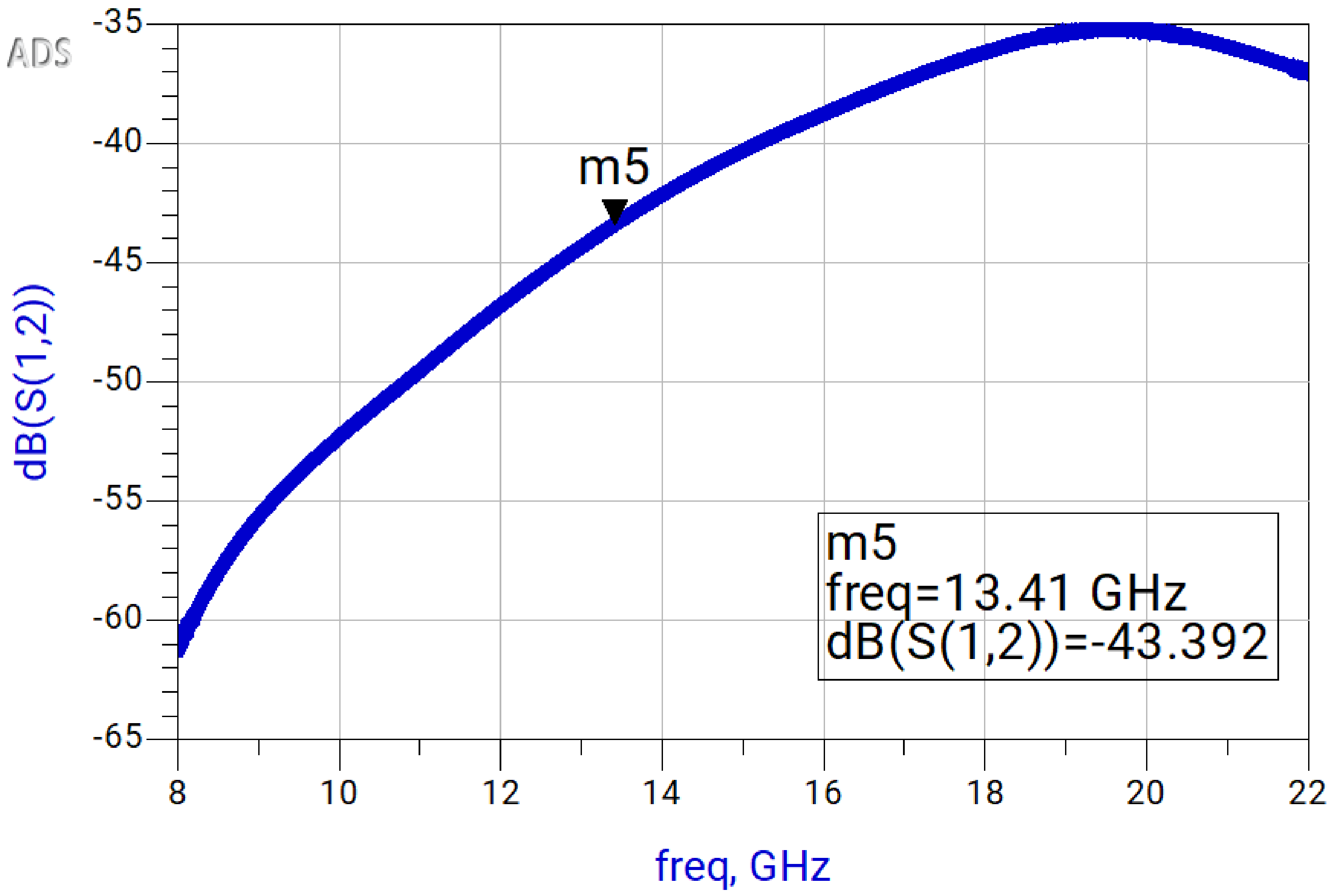
ADS Simulations



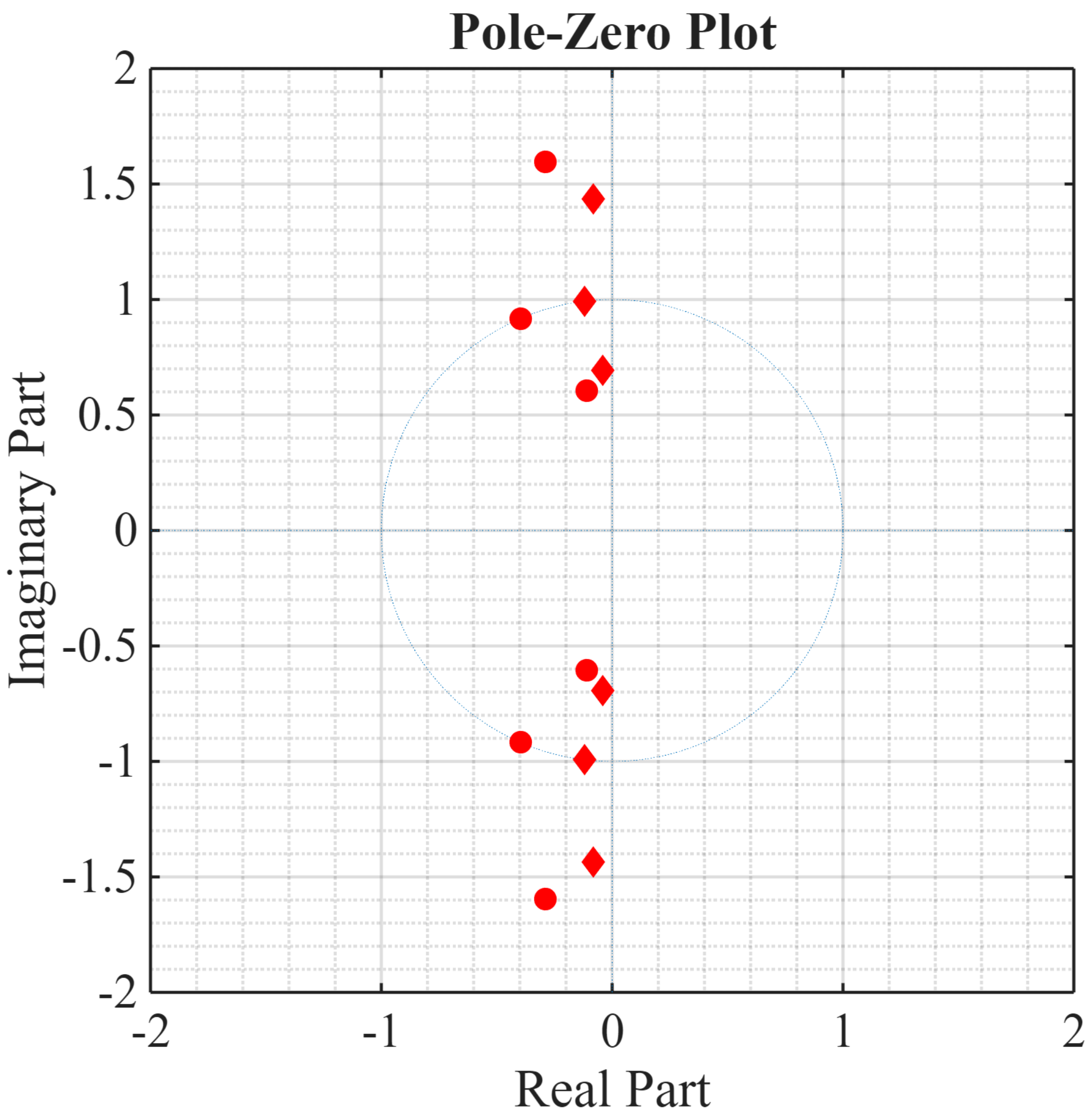
ADS Simulations



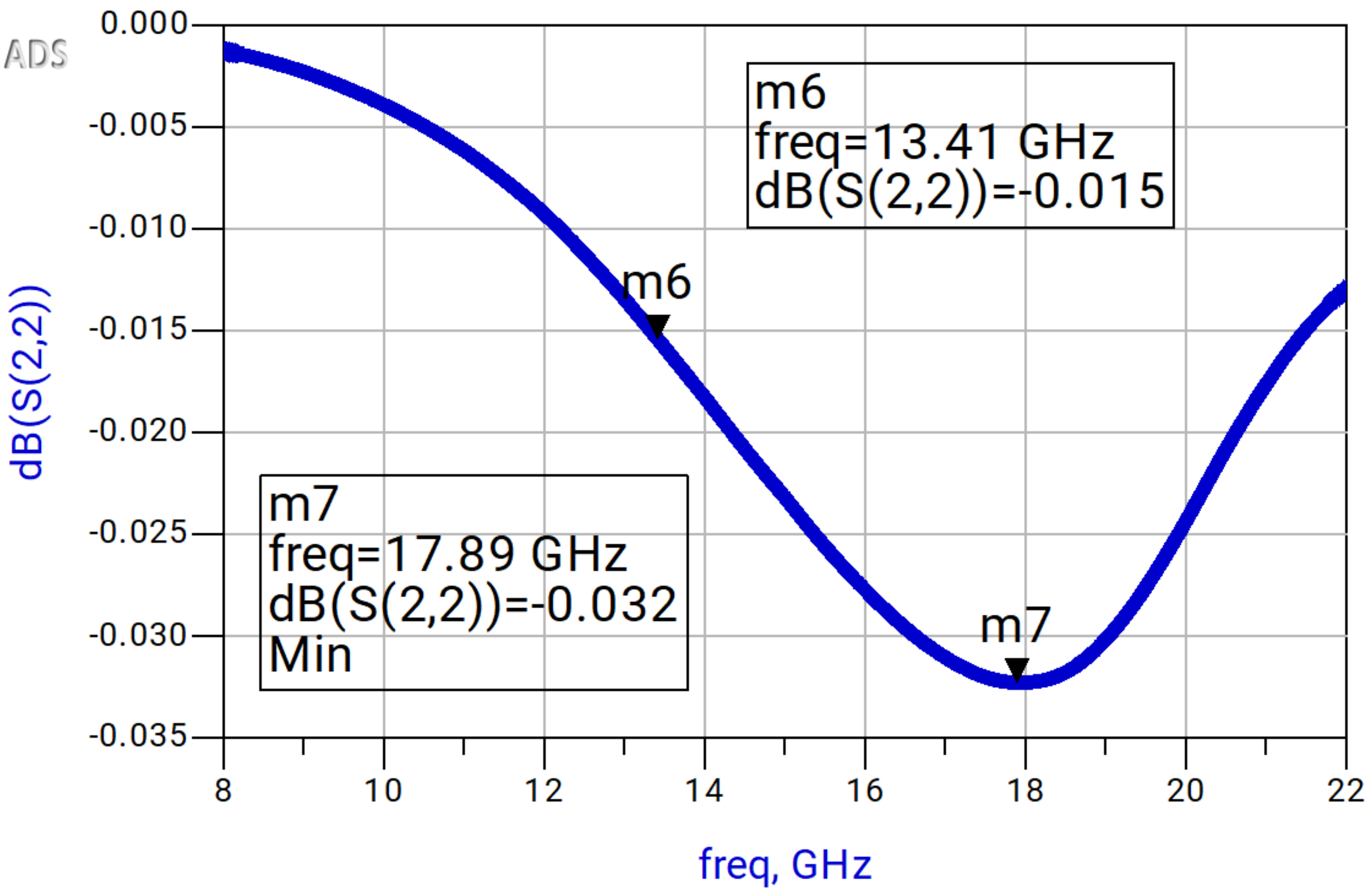
ADS Simulations



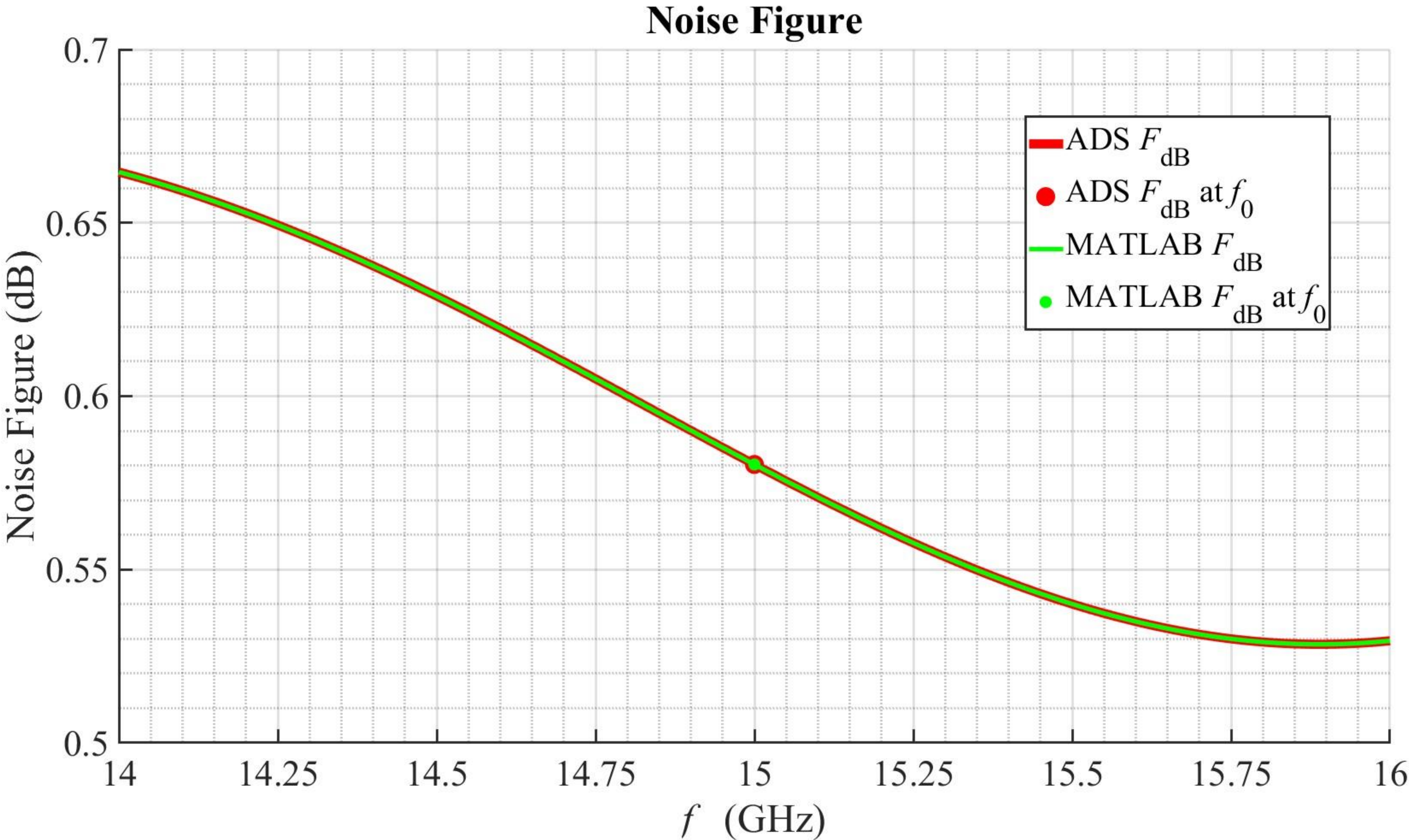
MATLAB Simulations



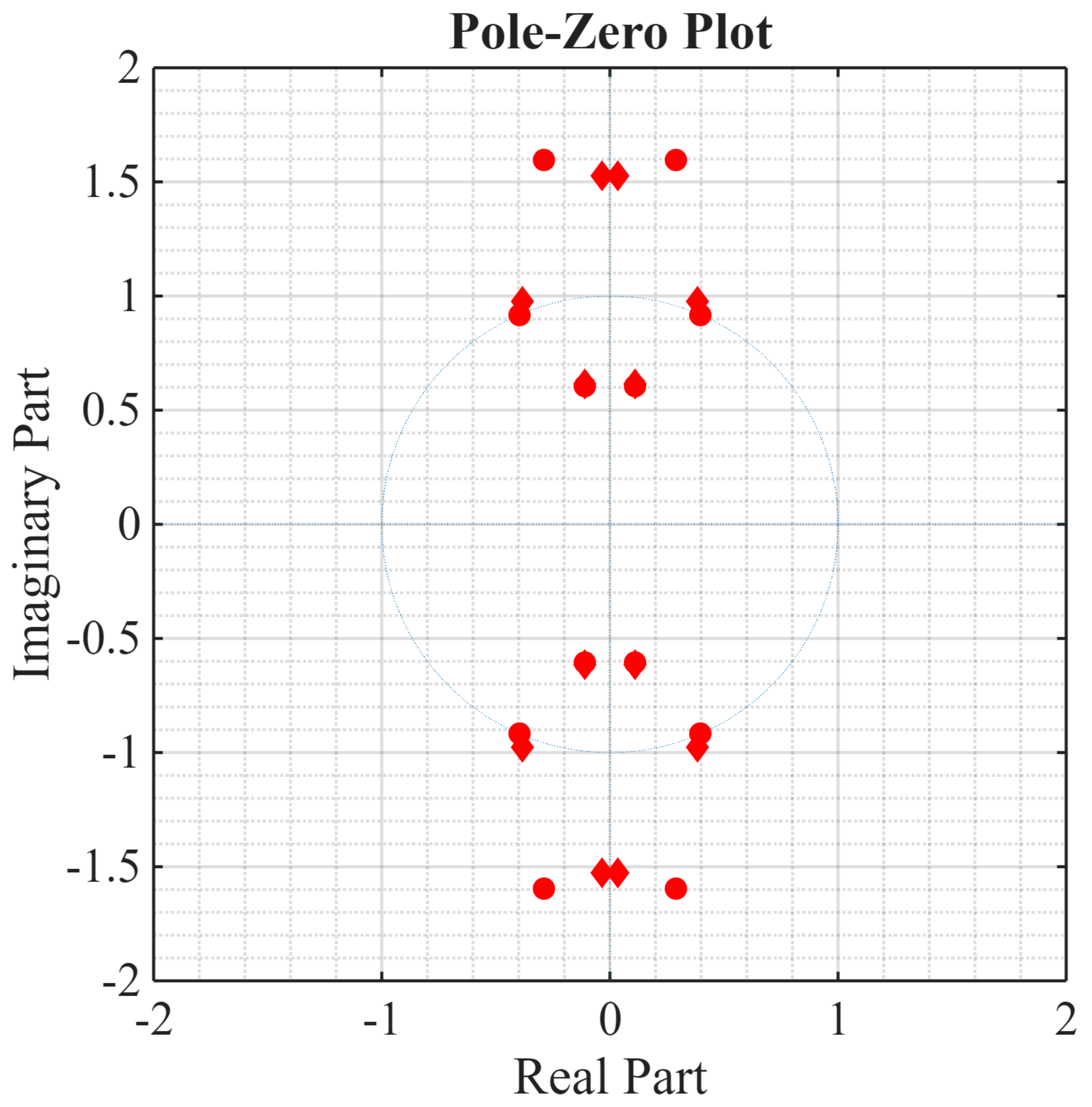
ADS Simulations



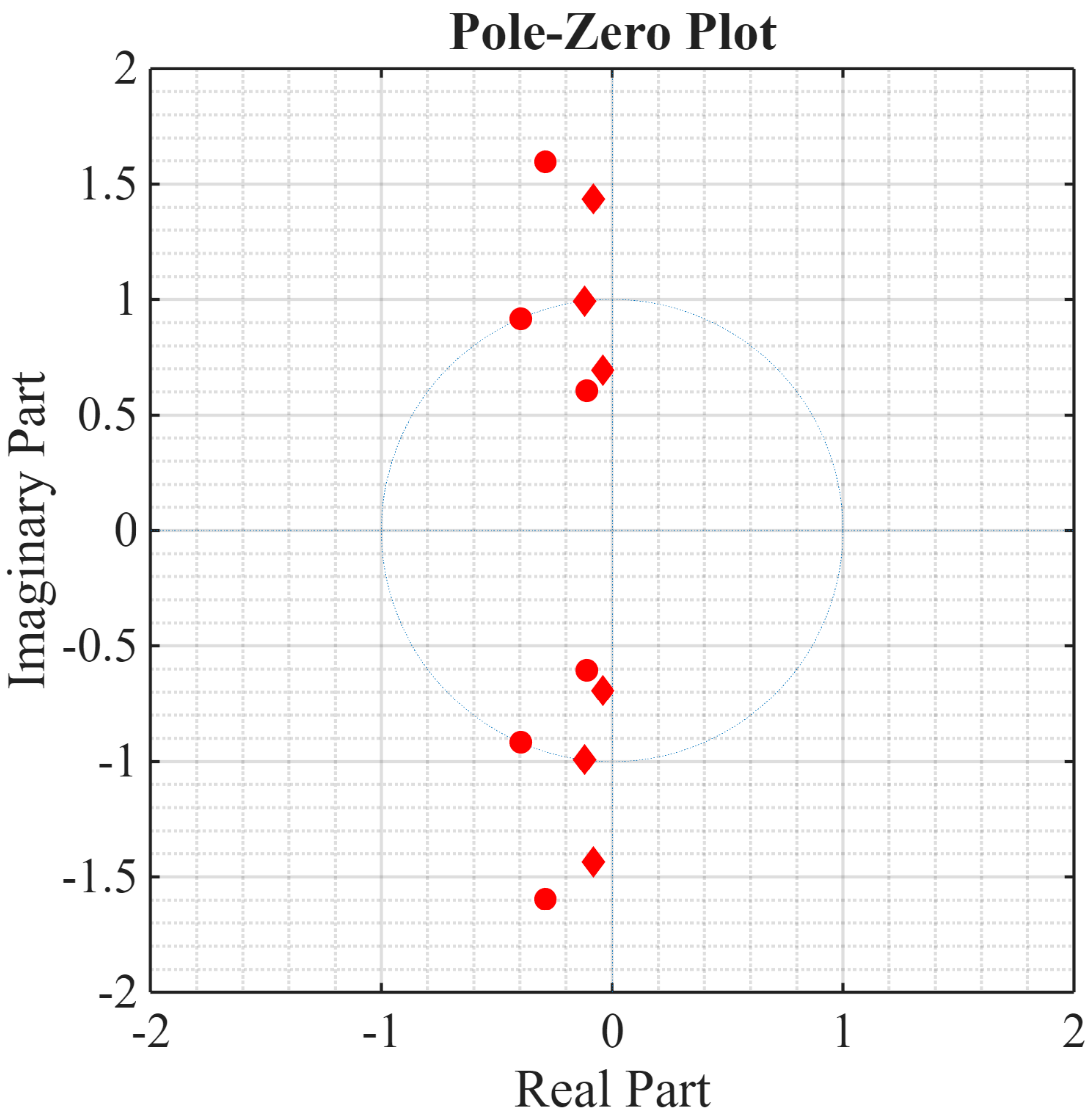
ADS & MATLAB Simulations



MATLAB Simulations



MATLAB Simulations



End of Design Project 3 Presentation

Additional Material Submitted to Kodiak:
- Excel
- Matlab Code
- ADS Files
- Scans / Screenshots

END

Appendix I — Excel Spreadsheets

	Design	Simulated Matlab	Simulated ADS	Units		ADS vs Matlab Diff	ADS vs Matlab % Diff
Gamma_S	(0.4256 , -136.0227°)	-----------------	-----------------	---		-----------------	-----------------
Gamma_L	(0.2904 , +146.0865°)	-----------------	-----------------	---		-----------------	-----------------
Gamma_In	(0.4742 , +123.6207°)	0.4688 , -10.5628°)	(0.5638 , +112.7047°)	---		-----------------	-----------------
Gamma_Out	(0.3642 , -168.6686°)	0.3555 , -25.0029°	(0.3642 , -168.6684°)	---		-----------------	-----------------
|Gamma_IMN|	0.1350	0.3031	0.3031	---		0.5803	0.0000
|Gamma_IMN|	-17.3944	-10.2445	-10.2445	dB		9.9673	0.0000
VSWR_IMN	1.4983	1.8879	1.8879	---		2.1651	0.0000
|Gamma_OMN|	0.1630	0.1629	0.1629	---		0.4401	0.0000
|Gamma_OMN|	-15.7583	-15.7583	-15.7583	dB		15.4811	0.0000
VSWR_OMN	1.3894	1.3894	1.3894	---		1.6666	0.0000
G_T	12.2599	12.1483	12.1483	dB		12.4255	0.0000
F	0.5803	0.5803	0.5803	dB		0.8575	0.0000

Simulation Comparison

This LNA configuration succeeded in meeting all the specifications which the design required.

Both the MATLAB and the ADS simulations provide the same values for the LNA.

MATLAB and ADS provided simulated values for the parameters listed in table 1,
these values are essentially the same, showing an excellent match:
— there is less than a 0.0001 percent difference between MATLAB and ADS values.

The noise figure could be lower, however overall gain provided by this design
isn't as lacking as many of the other designs solutions which were considered before this one.